1 Scope

The question is how the top venues of 2023 to 2026 state threat models for multi-party learning, and which way they are moving. The method is a screen of every accepted paper at eleven venues, followed by a reading of each kept paper and its coding into the ledger of the adversary interface, one row per coalition.

2 Result

The sweep screened 574 papers, placed 364, and coded the 350 that pass the inclusion rule into 519 ledger rows. Empirical: rows with a deviating adversary rose from 64 to 75 per cent between 2023 and 2026, and rows protected by HE rose from 12 to 26 per cent (Table 1). Empirical: among deviating-adversary rows whose evaluation says whether the adversary adapts to the defence, the adaptive share fell from 54 to 35 per cent (Table 2). The field therefore states a malicious adversary more often each year and evaluates it against an adaptive adversary less often.

Does not show: Each row comes from one coder's reading of the paper, so every note behind it is prov sweep, and the trend columns come from keyword matching on the coded cells. The 2026 counts lack NeurIPS 2026 and may lack part of CCS 2026. Screening recall is unmeasured, since the drop record lists only near misses. Citation data covers 117 papers so far, because the day's OpenAlex budget ran out, so no prominence ranking is given here. EuroS&P, SaTML and CSF were not screened.

3 How it was produced

Sources	Accepted-paper pages and proceedings of IEEE S&P, USENIX Security, CCS, NDSS, PoPETs, NeurIPS, ICML, ICLR, CRYPTO, EUROCRYPT and ASIACRYPT. DBLP was behind a bot check.
Range	2023 to 2026, with a few earlier anchors from the first sweeps
Screened	574 papers. 206 dropped at screening, 14 dropped after reading, 4 without an accessible PDF
Artifacts	`lib/literature/ledger.csv`, `lib/literature/screening.csv`, `lib/literature/notes/`, `sink/ledger-columns.md`
Reproduce	the trends script in `lib/reports/src/figures/`, run from the repository root

4 What this changes

The study's opening observation, that papers claim a malicious adversary and evaluate a weaker one, is now a measured trend over three years of top venues. The protection layer the group works in, HE, is the one gaining share. A next problem is therefore best placed where a deviating adversary meets an HE protection layer, and where evaluation against an adaptive adversary is rare. The ledger's edge and survives columns are the place to look for such a problem.

5 Method

Six screeners read the accepted-paper lists, one per venue group. Each kept a paper when it concerned training or fine-tuning by two or more parties holding data or models, and its contribution concerned privacy or security against an adversary. Each drop that needed an abstract to decide carries its reason in the screening record. PDFs came from the venues' open proceedings, from arXiv, and from the KU library. Every PDF was matched to its paper by the title and authors on its first page before it was placed with its note.

A coder then read each placed paper and wrote one row per coalition on the 22 columns of the column dictionary. A coder that found a paper outside the rule wrote no row, and the paper's note records the reason. Protocol papers on threshold or multiparty HE stay on the protocol axis with their learning columns set to n/a, because they bear on the group's own question. The merge takes

Table 1. Per cent of ledger rows per venue year, SoK rows excluded. Rows gives the count. Deviating rows set a node argument or output beyond queries. A mechanism column counts rows whose protection names it. Proven counts rows whose evidence status is proven or sketch only.

Year	Rows	Deviating	HE	DP	SecAgg	MPC	ZK	TEE	Proven
2023	127	64	12	41	21	20	10	3	76
2024	141	71	9	41	15	18	5	6	70
2025	132	74	17	42	23	23	14	7	80
2026	92	75	26	38	24	13	10	8	75

Table 2. Deviating-adversary rows by whether the evaluation includes an adversary that adapts to the defence. The share, in per cent, leaves out rows with adaptive set to n/a or unstated.

Year	Deviating rows	Adaptive	Not adaptive	Adaptive share
2023	81	28	24	54
2024	100	26	38	41
2025	98	25	29	46
2026	69	14	26	35

every venue from the paper's note, which carries the verified venue.